AN EXPLAINER FOR PHYSICISTS

Pinned ions, hard problems

Anyone who simulates ion crystals ends up annealing them hundreds of times and keeping the lowest result. Is that because the problem is fundamentally hard? The answer depends on one physical question: are the ions **free**, or **pinned**?

Paco Navarro · UC Santa Barbara · companion to the paper "When is finding the ground state of an ion Coulomb crystal NP-hard?"

The folklore

Load a hundred ions into a Paul trap and ask for their equilibrium structure. You cannot solve for it. You simulate, and because there are many metastable crystals with nearly the same energy, a single relaxation usually lands in the wrong one. Groups therefore anneal each parameter set hundreds of times, and some define the ground state operationally as the lowest configuration they recover at least ten times. The same story holds for the century-old Thomson problem of charges on a sphere, whose number of local minima is observed numerically to grow exponentially with the number of charges.

The usual explanation is that the problem is "NP-hard." We went looking for the proof. The Thomson problem is described in the literature as "known to be NP-hard," citing a 1985 paper. That paper proved hardness for particles on a lattice with site-dependent weights, and it was later shown not to apply to interactions that depend only on distance, like $1/r$. The closest real result is recent: Michael Emmerich proved in 2026 that choosing points of minimum energy in the plane is NP-hard for the inverse-square interaction $1/r^2$. For the Coulomb interaction of actual ions, as far as we can tell, nobody had proved it.

Complexity in a few minutes

Computer scientists sort problems by how the effort to solve them grows with the size of the input. Three ideas are enough here.

Checking versus finding. A problem is in **NP** if a proposed answer can be checked quickly. Hand me a crystal configuration and an energy target, and I can check it by summing $1/r$ over pairs. Finding a configuration that meets the target may still be very hard.

NP-hard means programmable. A problem is **NP-hard** if every problem in NP can be translated into it: scheduling, Sudoku, protein design, circuit satisfiability. Physicists met this in 1982, when Barahona showed that finding the ground state of a three-dimensional Ising spin glass is NP-hard. A fast spin-glass solver would be a fast solver for everything in NP, which almost everyone believes is impossible.

Proofs are constructions. To show a physical system is NP-hard, you build a translation, called a *reduction*, that turns any instance of a known hard puzzle into an instance of your system, so that solving the physics solves the puzzle.

> KEY IDEA
>
> NP-hardness is about programmability. A system is NP-hard when the problem you hand it carries enough adjustable structure to encode an arbitrary hard puzzle.

Surprise one: a free crystal cannot be NP-hard

Ask what the input to the free-crystal problem actually is. The trap is fixed, so the only thing that changes from one problem to the next is the ion number N. There is exactly one ground-state problem for each integer. That is far too little information to encode arbitrary puzzles.

The paper makes this precise. For any input size there are only polynomially many distinct questions, and their answers can be written down in advance as a list of polynomial length. Problems with that property form the class **P/poly**, and a classic theorem of Karp and Lipton says that if an NP-hard problem were in P/poly, the whole tower of complexity classes above NP (the polynomial hierarchy) would collapse. That is not believed to happen, so free crystals are not NP-hard. The argument never uses the Coulomb law, so it covers the Thomson problem too.

WHAT IS THE POLYNOMIAL HIERARCHY?

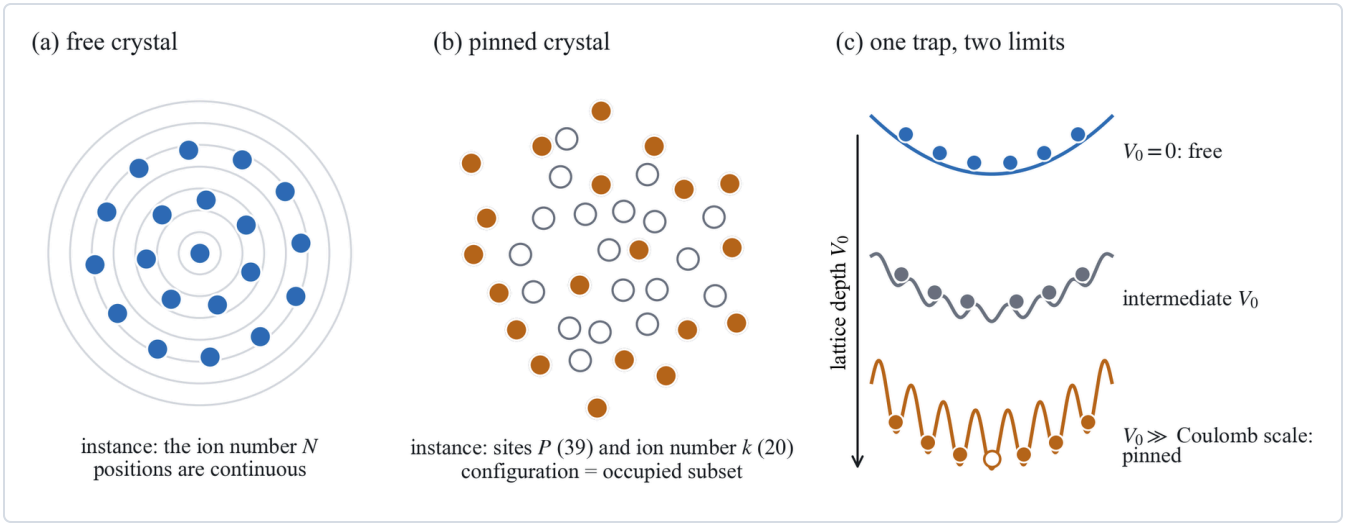
Complexity classes can be built from how many alternating "there exists" and "for all" questions a problem contains.

One "there exists" is NP: *is there* a configuration with energy below τ? A single example settles it.

One "for all" is the mirror image: *do all* configurations have energy above τ? No single example can prove that. This is why an annealing study can report the lowest crystal it found but cannot certify that nothing lower exists.

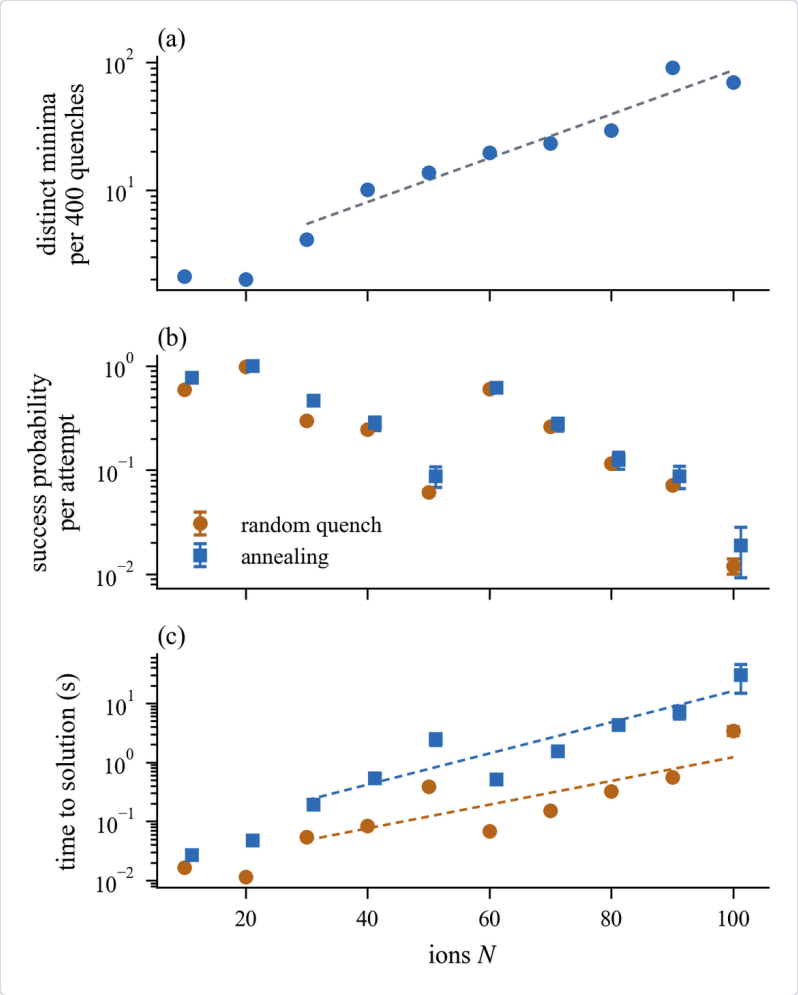
Two alternations combine them: *is there* a configuration such that *all* others lie above it? That is the full statement "this is the ground state." Three alternations look like a game: is there a move such that for every reply there is a winning counter-move? Each extra alternation is believed to give a genuinely harder class, and the infinite stack of these levels is the polynomial hierarchy.

A **collapse** would mean that above some level, adding more alternations stops buying any extra power, so the infinite tower is really only a few stories tall. Karp and Lipton showed that if an NP-hard problem were in P/poly, the tower would collapse to its second level. Nobody has proved that it doesn't collapse, but it is believed almost as firmly as P ≠ NP, which would flatten the tower completely. Calling free crystals NP-hard would therefore imply a dramatic, unexpected fact about computation itself.



Free versus pinned. (a) A free crystal: 20 ions settle where Coulomb repulsion and the trap place them, and the only input is the number 20. (b) A pinned crystal: a landscape of wells is prescribed, and the problem is which 20 of the 39 wells to fill. (c) Both are limits of one trap with a superimposed lattice of depth V_0.

None of this makes free crystals easy, as the next figure shows. We relaxed crystals of up to 100 ions in a two-dimensional trap from thousands of random starts. The number of distinct metastable crystals we find grows like $e^{N/25}$. A random relaxation finds the lowest crystal we know of about 1% of the time at $N = 100$, and the expected time to find it grows tenfold every 50 or so ions. Annealing helps per attempt but costs more per attempt, and its time to find the ground state grows exponentially as well.



Free crystals are hard in practice. (a) Distinct minima found per 400 relaxations. (b) Chance that one attempt finds the ground state. (c) Expected time to find it. Error bars are standard errors over 10 independent batches.

So the free crystal is hard in an exponential-search sense, with no structure for a clever algorithm to exploit. It is not hard in the NP sense, because there is nothing to program.

Surprise two: pin the ions and the problem becomes NP-hard

Now give the ions a landscape of wells, from a tweezer array or a segmented-electrode trap, deep enough that each ion sits at the bottom of a well. The ions can no longer move continuously. The only freedom left is *which* wells to fill. The well pattern is now part of the problem, and it can be chosen freely. That is the room needed to program.

Choosing which k wells to fill so as to minimize the Coulomb energy is NP-hard in two and three dimensions. If energies are only needed to finite precision, as in any real experiment or simulation, it is NP-complete: as hard as anything in NP, and no harder. The construction behind this is Emmerich's; the paper shows that it carries over from his inverse-square case to the Coulomb interaction.

The physics trick: hide bits in quadrupoles

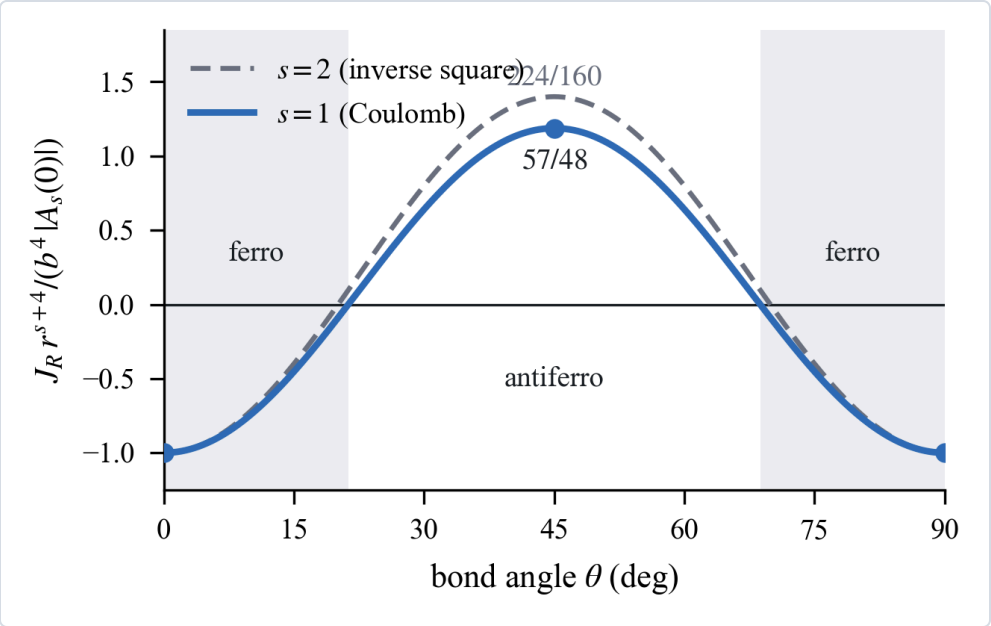
The obvious encoding fails. You might let “well occupied” mean 1 and rely on repulsion to keep occupied wells apart. But the Coulomb interaction is long range: every ion talks to every other ion, and the combined pull of all the distant pairs swamps any local logic you try to build.

Emmerich's fix is, at heart, a multipole argument. Build each bit from a tiny square of four wells holding two ions, and let the bit be which diagonal the ions sit on. Both choices have the same charge and the same (zero) dipole moment, and they differ only at quadrupole order. The interaction between two such bits therefore falls off as $1/r^5$ instead of $1/r$. The long-range Coulomb force becomes short-range logic.

The coupling between two squares separated by a vector at angle θ is

$$J \propto (9/2 - (105/2)\cos 4\theta)\, b^4/r^5,$$

where b is the size of the square. Along the lattice axes it is **ferromagnetic** (−48 in these units), so neighboring squares copy each other's bit and a chain of them acts as a wire. At 45° it is **antiferromagnetic** (+57), so two squares meeting there prefer to disagree.



The angular law. Coupling between two squares as a function of the angle between them, for the Coulomb case (solid) and the inverse-square case solved earlier (dashed). Shaded sectors copy bits; the 45° peak enforces "not both."

KEY IDEA

Encode information in rearrangements that leave the charge and dipole unchanged, not in occupation. Such a flip changes only the quadrupole, so distant squares feel it only weakly, as $1/r^5$, which turns an all-to-all interaction into a local circuit.

BUT DOESN'T THE COULOMB SUM DIVERGE?

It does. In a flat array, a ring of radius r holds about r squares, each contributing about $1/r$, so every ring adds a similar amount and the total energy grows without bound as the array gets larger. But only energy *differences* between arrangements decide the ground state, and the divergent part never enters them. Sorting the interaction between two squares by multipole order:

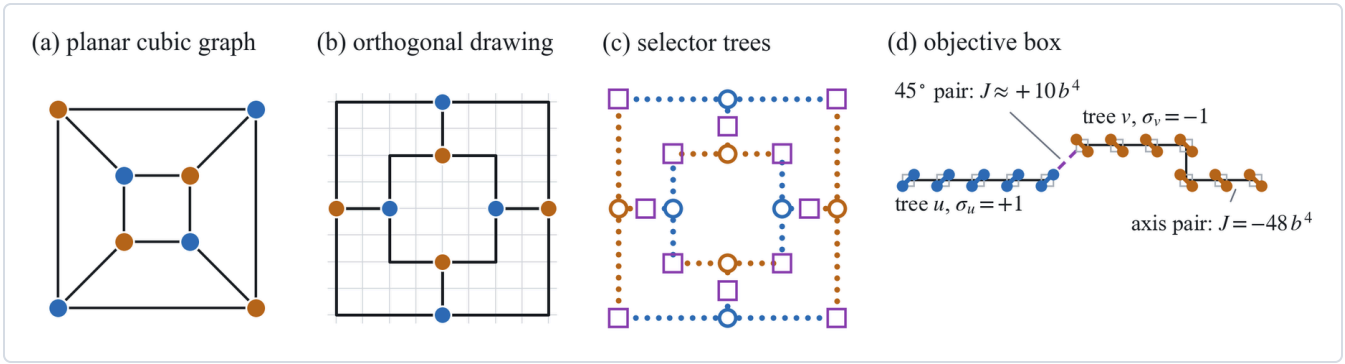
Charge ($1/r$). Every square holds two ions whichever diagonal they use, so this part is the same for every arrangement. It is a constant, like the weight of a ship you stand on while weighing a letter.

Dipole ($1/r^2$). Zero for both diagonals, so it drops out exactly.

Quadrupole against charge ($1/r^3$). A bias toward one diagonal. Rather than proving that the sum of these converges, the construction adds them up for each square and cancels them by stretching that square's diagonal very slightly.

Quadrupole against quadrupole ($1/r^5$). The coupling that does the logic. Sums of $1/r^5$ over a plane converge quickly, and gadgets that have nothing to do with each other are simply placed far apart.

With wires and "not both" junctions you can build a circuit for a classic NP-hard puzzle: given a network, pick as many nodes as possible with no two neighbors picked (a maximum independent set). Each node becomes a three-armed tree of wires, each link becomes a 45° junction, and a small bias makes "picked" favorable. The lowest-energy arrangement of the ions spells out the best answer.



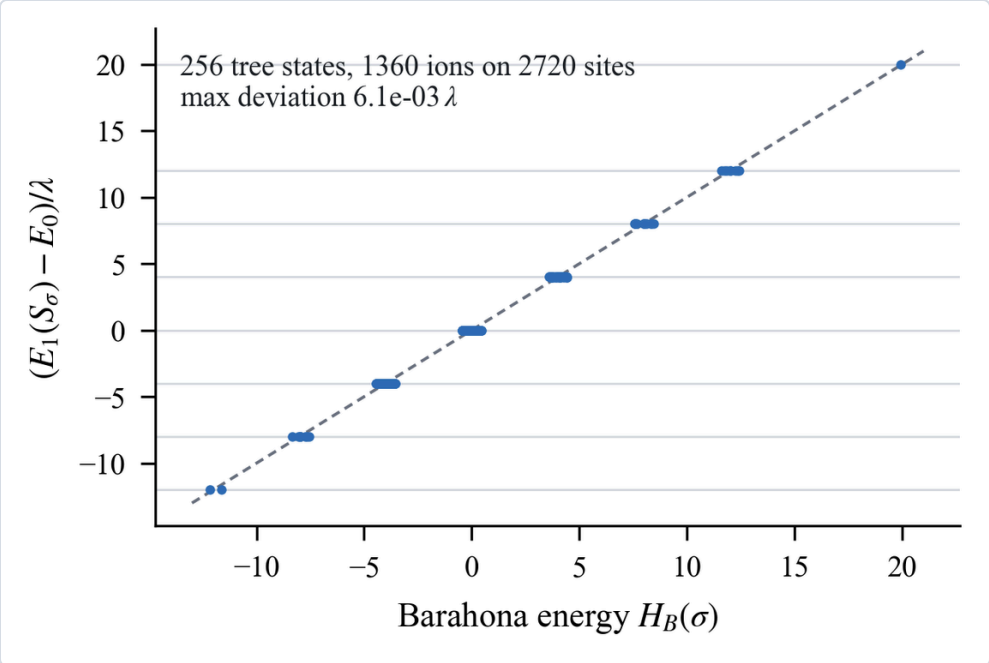
From puzzle to crystal. A network (a) is drawn on a grid (b). Each node becomes a tree of tiny squares and each link a 45° junction (c, d). Blue and orange are the two bit values.

All of this circuitry is Emmerich's, built for the inverse-square interaction. What the Coulomb case changes is the bookkeeping. The interaction decays one power more slowly

and the 45° coupling is relatively weaker, and two of the original estimates become too weak. One controls the side effects of the diagonal stretching; the other checks that a wire stays consistent. At the worst spot, the coupling that keeps a wire consistent beats everything working against it by 48 to 42, a 12% margin, which we confirmed with interval arithmetic. Bounding every pair by the strongest possible coupling, as is enough in the inverse-square case, would not work here.

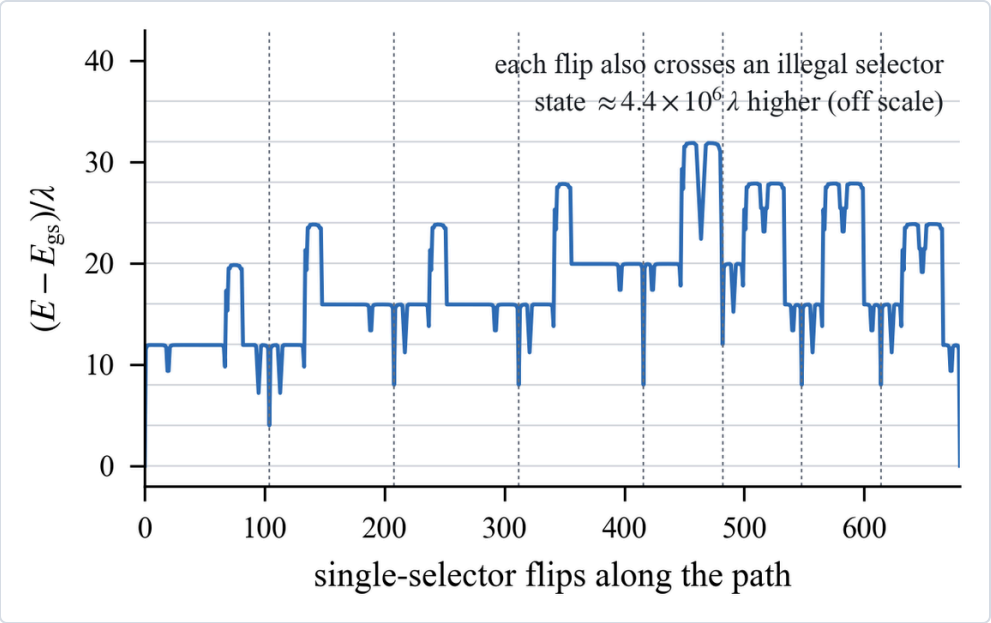
Seeing it work

We built the full instance for the cube network above: 1,360 ions on 2,720 wells. For all 256 logical states we computed the exact Coulomb energy, summing $1/r$ over every pair. The energies collapse onto evenly spaced levels, exactly as the puzzle's own scoring predicts, to better than 1% of the level spacing. The crystal really does compute the puzzle.



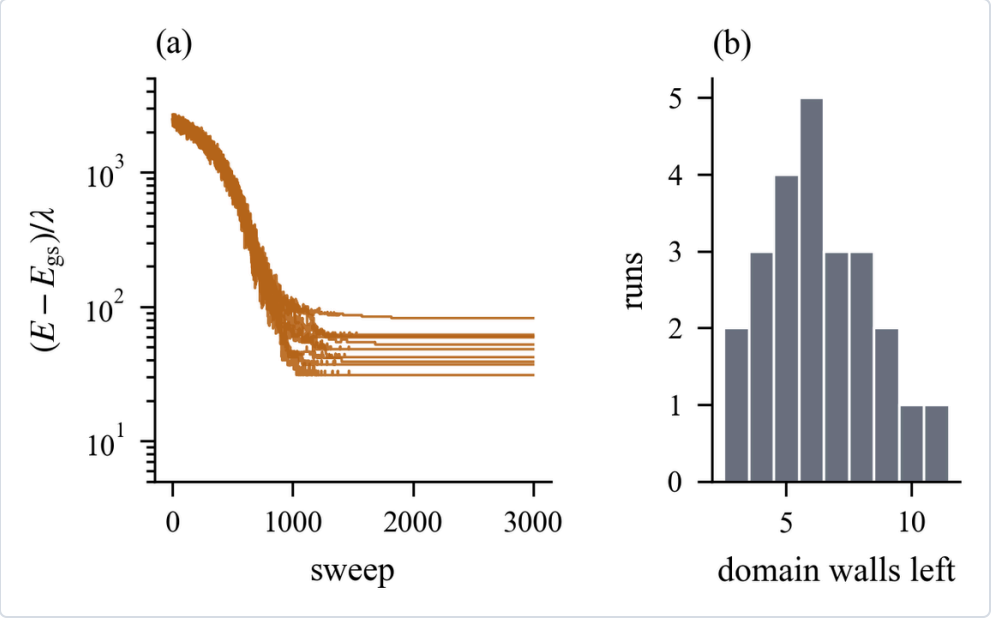
The crystal computes. Exact Coulomb energy of every logical state against the puzzle's score.

The resulting landscape looks like a spin glass. The two best answers are degenerate, but moving from one to the other means flipping whole trees, and the energy along the way climbs 32 times the level spacing as domain walls form and travel. Moving a single ion makes it far worse, roughly four million times the level spacing.



Barriers. Energy along a path of single-square flips connecting the two degenerate ground states.

Simulated annealing on this crystal behaves accordingly. None of 24 runs found the ground state. Every run froze with between 3 and 11 domain walls stuck inside the wires, even though the puzzle has only 256 logical answers.



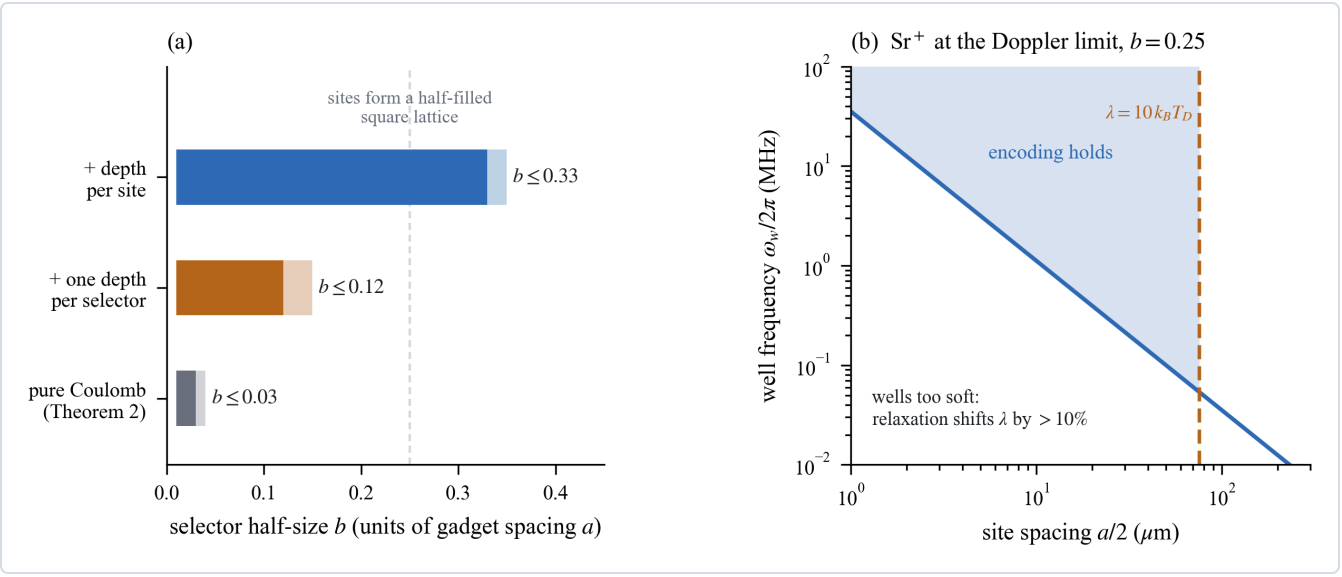
Annealing freezes. (a) Energy above the ground state during annealing. (b) Domain walls left at the end of each run.

Could you build one?

Not the proof's instances. To beat effects that grow with the size of the problem, the proof makes its squares absurdly small: as the network grows, the squares shrink like the inverse cube of the number of squares. No real wells are that close together. So we asked a more practical question. Take the cube-network crystal above, make the squares bigger, and check directly whether it still computes: does every square still hold two ions on a diagonal, do the wires still copy their bits, and do the energies still line up with the puzzle?

The answer depends on how much control you have over the wells. With identical wells, the encoding survives only while each square's half-width is about 3% of the spacing between squares. Beyond that, ions drift toward the edges of the array, the way charges spread out on a conductor. If you can set each well's depth, so that it cancels the average Coulomb potential at that well, the encoding survives up to a half-width of a third of the spacing. At a half-width of a quarter, the wells form a perfectly ordinary half-filled square lattice, so the logic runs in a geometry that looks like an ordinary crystal.

Then put in real ions. Doppler-cooled Sr^{+} sits at about half a millikelvin. With wells 20 μm apart, the puzzle's energy unit is about 40 times that thermal energy, and wells with a frequency of about 0.4 MHz are stiff enough that the ions' small shifts within their wells don't spoil the encoding. At 5 μm spacing the energy unit grows to about 150 times the thermal energy, but the wells must reach about 3 MHz. That is the regime of electrode trap arrays, not optical lattices, whose wells are far too shallow to hold ions against their neighbors.



Toward the lab. (a) How large the squares' half-width can be, relative to their spacing, before the cube-network crystal stops computing, for three levels of control over the wells. (b) For Sr^+ at the Doppler limit with square half-width a quarter of their spacing, the shaded region is where the wells are stiff enough and the energy unit is at least ten times the thermal energy.

Two cautions. This is one instance, so it shows where the mechanism works, not a proof that hardness survives at realistic spacing. And being free to set every well's depth is itself a powerful knob: with arbitrary site-dependent energies, hardness is much less surprising.

At a glance

	FREE CRYSTAL	PINNED CRYSTAL
You specify	the number of ions	the well landscape and the number of ions
Configuration	continuous positions	which wells are filled
NP-hard?	no, unless the polynomial hierarchy collapses	yes, in 2D and 3D
In practice	exponential search, about 10× per 50 ions	glassy: near-degenerate states behind large barriers
In the lab	any trap, any cooling	stiff wells with tunable depths, a few to tens of μm apart

Why it matters, and what is open

For experiments with deep pinning potentials, the result says no algorithm can be guaranteed to find the optimal loading pattern efficiently, so heuristics are unavoidable. Mixed-species crystals raise related questions. In single-molecule spectroscopy, one molecular ion sits among laser-cooled atomic ions and is read out through the crystal's motion, and where it sits is itself an optimization over many nearly degenerate configurations.

Several questions are open, and some are physics questions in disguise.

- **Is there a complexity transition?** Free crystals are not NP-hard and deeply pinned ones are. What happens at intermediate lattice depth, where ions can partly leave their wells?
- **Is a realistic lattice enough?** The numerics suggest the encoding works on a half-filled square lattice of wells. Proving hardness there, with wells no closer than a fixed spacing, is open.
- **How many ground states?** Counting them is a question about zero-temperature entropy, and its complexity is unknown.
- **Exactly how hard?** Checking an answer exactly means comparing sums of square roots, a famous unsolved problem in its own right.

The logical core of the proof has been checked by machine, in the Lean proof assistant. The geometric estimates are proved by hand, and the tightest of them, the lattice sums behind the 12% margin, are certified by interval arithmetic. All the code is in the accompanying repository.

Glossary

NP	Problems whose proposed answers can be checked quickly.
NP-hard	At least as hard as every problem in NP; any of them can be translated into it.
NP-complete	Both in NP and NP-hard: the hardest problems whose answers are easy to check.

Reduction	An efficient translation from one problem into another that preserves the answer.
Polynomial hierarchy	The tower of classes built from alternating “there exists” and “for all” questions; NP is its first level. A collapse means the tower is only finitely tall.
P/poly	Problems solvable quickly with the help of a precomputed list whose length grows only polynomially with input size.
Time to solution	Expected time to reach the ground state with a randomized method: cost per attempt divided by the success probability per attempt.

P. Navarro, “When is finding the ground state of an ion Coulomb crystal NP-hard?”, manuscript in preparation (2026). The construction it builds on is M. T. M. Emmerich, “Riesz energy subset selection in the Euclidean plane is NP-hard: A reduction from the Ising model on planar cubic graphs,” arXiv:2608.23506 (2026).